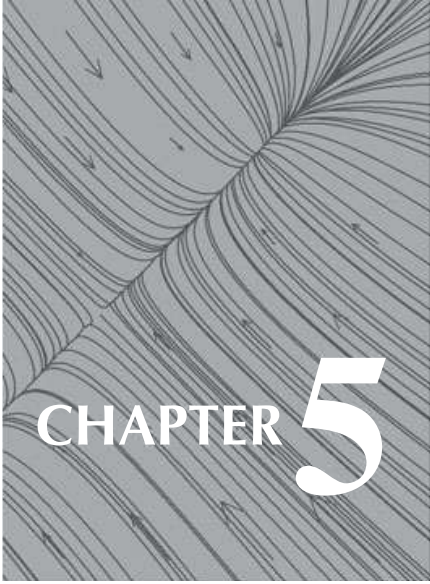




CHAPTER 5

Optimal Control

BEFORE me grew the human soul,
And after I am dead and gone,
Through grades of effort and control
The marvellous work shall still go on

—Archibald Lampman

If everything seems under control, you're just not going fast enough.

—Mario Andretti

One of the essential elements of the control problem, discussed in Section 1.1, is a means of testing the performance of any proposed control law. Whenever we use the term “best” or “optimal” to describe the effectiveness of a given control strategy, we do so with respect to some numerical index of performance called the performance index. We assume that the value of the performance index decreases as the quality of the given admissible control law increases. The admissible controller that ensures the completion of the system objective and at the same time minimizes the performance index is called an *optimal controller* for the system. In this chapter, we discuss various methods of constructing optimal controllers.

5.1 Performance Indices

Constructing a performance index—that is, choosing a means to measure the system performance—can be considered a part of the system modeling. Here, we discuss some typical choices of performance indices.

First, suppose that the objective is to control a dynamical system modeled by the equations

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t), \quad \mathbf{x}(t_0) = \mathbf{x}_0, \quad (5.1)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) \quad (5.2)$$

on a fixed interval $[t_0, t_f]$ so that the components of the state vector are “small.” A suitable performance index to be minimized would be

$$J_1 = \int_{t_0}^{t_f} \mathbf{x}^T(t)\mathbf{x}(t) dt. \quad (5.3)$$

Obviously, if J_1 is small, then the state vector norm, $\|\mathbf{x}(t)\|$, is small in the sense of the above performance index.

If the objective is to control the system so that the components of the output, $\mathbf{y}(t)$, are to be small, then we could use the performance index

$$\begin{aligned} J_2 &= \int_{t_0}^{t_f} \mathbf{y}^T(t) \mathbf{y}(t) dt \\ &= \int_{t_0}^{t_f} \mathbf{x}^T(t) \mathbf{C}^T \mathbf{C} \mathbf{x}(t) dt \\ &= \int_{t_0}^{t_f} \mathbf{x}^T(t) \mathbf{Q} \mathbf{x}(t) dt, \end{aligned} \quad (5.4)$$

where the weight matrix $\mathbf{Q} = \mathbf{C}^T \mathbf{C}$ is symmetric positive semidefinite.

In the case when we wish to control the system in such a manner that the components of the input, $\mathbf{u}(t)$, are “not too large,” a suitable performance index to be minimized is

$$J_3 = \int_{t_0}^{t_f} \mathbf{u}^T(t) \mathbf{u}(t) dt, \quad (5.5)$$

or

$$J_4 = \int_{t_0}^{t_f} \mathbf{u}^T(t) \mathbf{R} \mathbf{u}(t) dt, \quad (5.6)$$

where the weight matrix $\mathbf{R}$ is symmetric positive definite. There is no loss of generality in assuming the weight matrix $\mathbf{R}$ to be symmetric. If $\mathbf{R}$ was not symmetric, we could represent the quadratic term $\mathbf{u}^T \mathbf{R} \mathbf{u}$ equivalently as

$$\mathbf{u}^T \mathbf{R} \mathbf{u} = \mathbf{u}^T \left(\frac{\mathbf{R} + \mathbf{R}^T}{2} \right) \mathbf{u},$$

where the matrix $\frac{1}{2}(\mathbf{R} + \mathbf{R}^T)$ is symmetric.

As pointed out by Owens [218, p. 186], we cannot simultaneously minimize the performance indices (5.3) and (5.5) because minimization of (5.3) requires large control signals, while minimization of (5.5) requires small control signals. To solve the dilemma, we could compromise between the two conflicting objectives by minimizing the performance index that is a convex combination of J_1 and J_3 ,

$$\begin{aligned} J &= \lambda J_1 + (1 - \lambda) J_3 \\ &= \int_{t_0}^{t_f} (\lambda \mathbf{x}^T(t) \mathbf{x}(t) + (1 - \lambda) \mathbf{u}^T(t) \mathbf{u}(t)) dt, \end{aligned} \quad (5.7)$$

where λ is a parameter in the range $[0, 1]$. If $\lambda = 1$, then $J = J_1$; and if $\lambda = 0$, then $J = J_3$. By trial and error, we could select λ from the interval $[0, 1]$ to compromise between the two extremes. A generalization of the performance index (5.7) is

$$J = \int_{t_0}^{t_f} (\mathbf{x}^T(t) \mathbf{Q} \mathbf{x}(t) + \mathbf{u}^T(t) \mathbf{R} \mathbf{u}(t)) dt.$$

In certain applications, we may wish the final state $\mathbf{x}(t_f)$ to be as close as possible to $\mathbf{0}$. Then, a possible performance measure to be minimized is

$$\mathbf{x}^T(t_f) \mathbf{F} \mathbf{x}(t_f), \quad (5.8)$$

where $\mathbf{F}$ is a symmetric positive definite matrix.

We can combine the performance measures (5.4), (5.6), and (5.8) when our control aim is to keep the state “small,” the control “not too large,” and the final state as near to $\mathbf{0}$ as possible. The resulting performance index is

$$J = \frac{1}{2} \mathbf{x}^T(t_f) \mathbf{F} \mathbf{x}(t_f) + \frac{1}{2} \int_{t_0}^{t_f} (\mathbf{x}^T(t) \mathbf{Q} \mathbf{x}(t) + \mathbf{u}^T(t) \mathbf{R} \mathbf{u}(t)) dt, \quad (5.9)$$

where the factor $\frac{1}{2}$ is placed to simplify subsequent algebraic manipulations. Minimizing (5.9) subject to (5.1) is called the *linear quadratic regulator* problem, or the LQR problem for short.

In some cases, the controller’s goal is to force the system state to track a desired state trajectory, $\mathbf{x}_d(t)$, throughout the interval $[t_0, t_f]$ while maintaining the deviations of the actual state $\mathbf{x}(t)$ “small” from the desired trajectory with the control effort $\mathbf{u}(t)$ “not too large” and the final state $\mathbf{x}(t_f)$ being as near as possible to some desired state $\mathbf{x}_d(t_f)$. A suitable performance index to be minimized in this case might be

$$\begin{aligned} J = & \frac{1}{2} (\mathbf{x}(t_f) - \mathbf{x}_d(t_f))^T \mathbf{F} (\mathbf{x}(t_f) - \mathbf{x}_d(t_f)) \\ & + \frac{1}{2} \int_{t_0}^{t_f} ((\mathbf{x}(t) - \mathbf{x}_d(t))^T \mathbf{Q} (\mathbf{x}(t) - \mathbf{x}_d(t)) + \mathbf{u}^T(t) \mathbf{R} \mathbf{u}(t)) dt. \end{aligned} \quad (5.10)$$

5.2 A Glimpse at the Calculus of Variations

The goal of this section is to provide a brief introduction to the subject of the *calculus of variations*. The calculus of variations deals with minimization or maximization of *functionals*. A functional is a mapping or a transformation that depends on one or more functions and the values of the functional are numbers. Examples of functionals are performance indices that we introduced in the previous section. We now give two more examples of functionals.

◆ Example 5.1

The length l of the arc connecting two given points $A(x_0, y_0)$ and $B(x_1, y_1)$ in the x - y plane, as depicted in Figure 5.1, is an example of a functional. We know that

$$l = l(y(x)) = \int_{x_0}^{x_1} \sqrt{1 + (dy(x)/dx)^2} dx,$$

subject to the constraints $y(x_0) = y_0$ and $y(x_1) = y_1$.

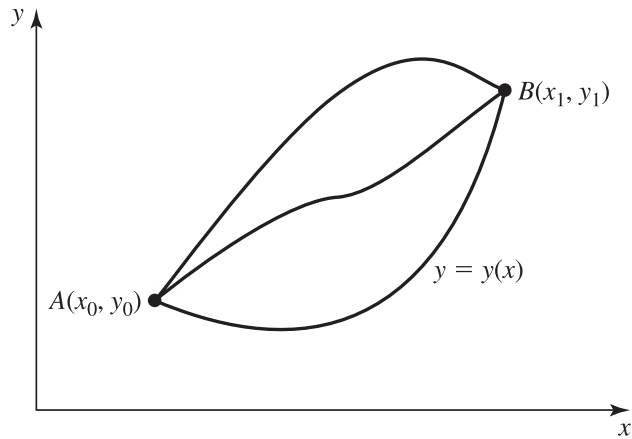


Figure 5.1 The length of arcs joining two given points in the plane is an example of a functional.

◆ Example 5.2

Another example of a functional is the area of the surface of revolution formed by a surface connecting two circular disks as illustrated in Figure 5.2. In particular, the surface of revolution can be formed by a soap film clinging to two circular disks. When a wire circle is dipped into a soap solution and withdrawn, a circular disk of soap film bounded by the circle is formed. If a second smaller circle is made to touch the disk and then moved away, the two circles will be joined by a surface of film that is a surface of revolution in the particular case when the circles are parallel and have their centers on the same axis perpendicular to their planes. If we denote the arc of intersection with the x - y plane by $y = y(x)$, then the area of the surface of revolution is

$$v = v(y(x)) = 2\pi \int_{x_0}^{x_1} y(x) \sqrt{1 + (dy(x)/dx)^2} dx,$$

subject to the constraints $y(x_0) = y_0$, $y(x_1) = y_1$. One may then want to minimize $v = v(y(x))$ to find $y = y(x)$ resulting in the surface of revolution of minimum area.

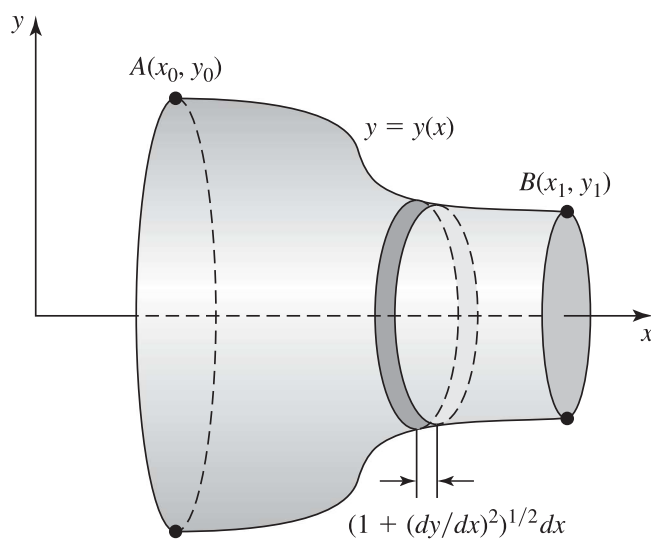


Figure 5.2 The area of surfaces of revolution connecting two circular disks is an example of a functional.

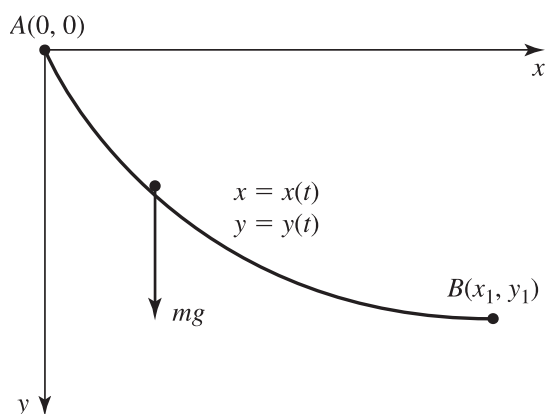


Figure 5.3 Illustration of the brachistochrone problem.

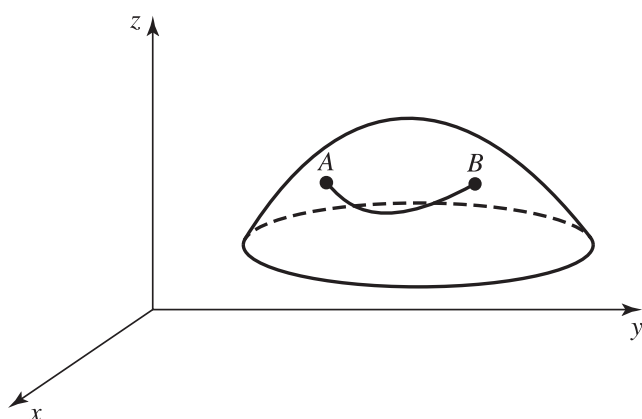


Figure 5.4 The problem of finding geodesics.

The problem of optimization of functionals dates back to Johann Bernoulli. In 1696 he published a letter in which he posed the problem, which is now known as the *brachistochrone problem*. Brachistochrone translates from Greek as the shortest time. The brachistochrone problem can be described as follows. A particle of mass m is to slide down along a frictionless track connecting two given points. The motion of the particle is to be constrained only by gravity and the shape of the curve of the track. Find the curve of the track that minimizes the time travel between the two given points. We illustrate the problem in Figure 5.3. The brachistochrone problem was solved by Johann Bernoulli himself, his brother Jacob, Isaac Newton, the Marquis de L'Hôpital, Gottfried Leibniz, and Tschirnhaus. A solution to the brachistochrone problem is presented in Example 5.23. In 1697 Johann Bernoulli posed and solved the *geodesics* problem, which can be formulated as follows: Find a line of shortest length that connects two points on a given surface; see Figure 5.4 for an illustration of the problem. The curves of the least length connecting pairs of points on a surface are called *geodesics*. A general procedure for solving the above type of problems, using the calculus of variations, was given by Leonhard Euler (who flourished between 1707 and 1783).

To acquaint ourselves with the methods of the calculus of variations, we first need to introduce concepts concerning functionals.

5.2.1 Variation and Its Properties

A *function* can be viewed as a rule, or a mapping, that assigns to each element of some set a unique element of a possibly different set. In particular, a function $x : \mathbb{R} \rightarrow \mathbb{R}$ of a real variable t

assigns to each real number a unique real number. An increment of the argument of a function of one variable t is $\Delta t = t - t_1$.

Similarly, a *functional* is a mapping that assigns to each function, from some class of functions, a unique number. Thus, we can say that a functional is a “function of a function.” Let $x : \mathbb{R} \rightarrow \mathbb{R}$ be an argument of a functional. By a *variation* δx of an argument x of a functional v we mean the difference of two functions

$$\delta x = x - x_1.$$

We assume that x can change in an arbitrary way in some class of functions.

Recall that a function $x : \mathbb{R} \rightarrow \mathbb{R}$ is continuous if a small change of its argument t corresponds to a small change of the value $x(t)$ of the function. Similarly, a functional v is said to be continuous if a “small” change of its argument x corresponds to a small change of the value of the functional.

A functional v is called linear if

$$v(ax_1 + x_2) = av(x_1) + v(x_2),$$

where a is a constant.

We will now introduce the notion of the *variation of a functional*. This notion is analogous to the notion of a function differential. To connect the two, we first consider the case of a function of one variable. Let x be a differentiable function defined on an open interval U , and let $t \in U$. The derivative of x at t is defined as

$$x'(t) = \lim_{\Delta t \rightarrow 0} \frac{x(t + \Delta t) - x(t)}{\Delta t}. \quad (5.11)$$

Let

$$\varphi(\Delta t) = \frac{x(t + \Delta t) - x(t)}{\Delta t} - x'(t). \quad (5.12)$$

The function φ is not defined at $\Delta t = 0$; however,

$$\lim_{\Delta t \rightarrow 0} \varphi(\Delta t) = 0.$$

We rewrite (5.12) as

$$x(t + \Delta t) - x(t) = x'(t)\Delta t + \varphi(\Delta t)\Delta t. \quad (5.13)$$

The above relation has meaning only when $\Delta t \neq 0$. To make it hold at $\Delta t = 0$, we define $\varphi(\Delta t)|_{\Delta t=0} = 0$. To proceed, let

$$\begin{aligned} \beta(\Delta t) &= \varphi(\Delta t) & \text{if } \Delta t > 0 \\ \beta(\Delta t) &= -\varphi(\Delta t) & \text{if } \Delta t < 0. \end{aligned} \quad (5.14)$$

Then, if x is differentiable, there exists a function β such that

$$\begin{aligned} x(t + \Delta t) - x(t) &= \Delta x = x'(t)\Delta t + \beta(\Delta t)|\Delta t| \\ &= L(t, \Delta t) + \beta(\Delta t)|\Delta t|, \end{aligned} \quad (5.15)$$

where $\lim_{\Delta t \rightarrow 0} \beta(\Delta t) = 0$, and $L(t, \Delta t) = x'(t)\Delta t$ is a linear function in Δt .

For a real-valued function $f = f(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}$, relation (5.15) takes the form

$$\begin{aligned} f(\mathbf{x} + \Delta \mathbf{x}) - f(\mathbf{x}) &= \Delta f = Df(\mathbf{x})\Delta \mathbf{x} + \beta(\Delta \mathbf{x})\|\Delta \mathbf{x}\| \\ &= L(\mathbf{x}, \Delta \mathbf{x}) + \beta(\Delta \mathbf{x})\|\Delta \mathbf{x}\|, \end{aligned} \quad (5.16)$$

where $\lim_{\Delta \mathbf{x} \rightarrow \mathbf{0}} \beta(\Delta \mathbf{x}) = 0$, and

$$L(\mathbf{x}, \Delta \mathbf{x}) = Df(\mathbf{x}) \Delta \mathbf{x} = \nabla f(\mathbf{x})^T \Delta \mathbf{x}$$

is a linear function in $\Delta \mathbf{x}$. Relation (5.16) holds for operators defined over Banach spaces with scalars being either real or complex numbers. The functionals we consider are operators over a Banach space of continuous functions, $\mathcal{C}([t_0, t_1])$, with the norm given by (A.99) on page 656.

If an increment $\Delta v = v(x + \delta x) - v(x)$ of a functional v can be represented as

$$\Delta v = L(x, \delta x) + \beta(\delta x) \|\delta x\|,$$

where $L(x, \delta x)$ is a linear functional with respect to δx , the term $\|\delta x\| = \max_{t \in [t_0, t_1]} |\delta x|$ denotes the maximal value of $|\delta x|$, and $\beta(\delta x) \rightarrow 0$ if $\|\delta x\| \rightarrow 0$, then the linear part L of Δv is called the *variation of the functional* and is denoted δv ; that is,

$$\delta v = L(x, \delta x).$$

◆ Example 5.3

We find the variation of the functional

$$v = \int_0^1 (2x^2(t) + x(t)) dt.$$

For this we first calculate its increment to get

$$\begin{aligned} \Delta v &= v(x + \delta x) - v(x) \\ &= \int_0^1 (2(x + \delta x)^2 + (x + \delta x)) dt - \int_0^1 (2x^2 + x) dt \\ &= \int_0^1 (2x^2 + 4x\delta x + 2(\delta x)^2 + x + \delta x - 2x^2 - x) dt \\ &= \int_0^1 (4x + 1)\delta x dt + 2 \int_0^1 (\delta x)^2 dt. \end{aligned}$$

The linear part of Δv is

$$\delta v = \int_0^1 (4x + 1)\delta x dt,$$

which is the variation of the given functional.

To proceed, note that the linear part on the right-hand side of (5.16) can be computed as

$$Df(\mathbf{x}) \Delta \mathbf{x} = L(\mathbf{x}, \Delta \mathbf{x}) = \left. \frac{d}{d\alpha} f(\mathbf{x} + \alpha \Delta \mathbf{x}) \right|_{\alpha=0} \quad (5.17)$$

As mentioned before, the above also holds for operators over Banach spaces other than Euclidean

spaces. Using this observation, we can state and prove a lemma that will allow us to calculate the variation of a given functional more efficiently than the method we used in Example 5.3.

Lemma 5.1

$$\delta v = \left. \frac{d}{d\alpha} v(x + \alpha \delta x) \right|_{\alpha=0}.$$

Proof Suppose that for a given functional v there exists its variation. This means that we can represent Δv as

$$\Delta v = v(x + \alpha \delta x) - v(x) = L(x, \alpha \delta x) + \beta(\alpha \delta x) |\alpha| \|\delta x\|.$$

Then, the derivative of $v(x + \alpha \delta x)$ with respect to α evaluated at $\alpha = 0$ is equal to

$$\begin{aligned} \lim_{\alpha \rightarrow 0} \frac{\Delta v}{\alpha} &= \lim_{\alpha \rightarrow 0} \frac{L(x, \alpha \delta x) + \beta(\alpha \delta x) |\alpha| \|\delta x\|}{\alpha} \\ &= \lim_{\alpha \rightarrow 0} \frac{L(x, \alpha \delta x)}{\alpha} + \lim_{\alpha \rightarrow 0} \frac{\beta(\alpha \delta x) |\alpha| \|\delta x\|}{\alpha} \\ &= L(x, \delta x), \end{aligned}$$

because $L(\cdot, \cdot)$ is linear with respect to the second argument, and hence

$$L(x, \alpha \delta x) = \alpha L(x, \delta x).$$

Furthermore,

$$\lim_{\alpha \rightarrow 0} \frac{\beta(\alpha \delta x) |\alpha| \|\delta x\|}{\alpha} = \lim_{\alpha \rightarrow 0} \beta(\alpha \delta x) \|\delta x\| = 0.$$

This completes the proof.

◆ Example 5.4

We will now use Lemma 5.1 to find the variation of the functional considered in Example 5.3. We have

$$\begin{aligned} \delta v &= \left. \frac{d}{d\alpha} v(x + \alpha \delta x) \right|_{\alpha=0} \\ &= \left. \frac{d}{d\alpha} \left(\int_0^1 (2(x + \alpha \delta x)^2 + (x + \alpha \delta x)) dt \right) \right|_{\alpha=0} \\ &= \left. \int_0^1 (4(x + \alpha \delta x) \delta x + \delta x) dt \right|_{\alpha=0} \\ &= \int_0^1 (4x + 1) \delta x dt, \end{aligned}$$

which is the variation of the given functional in Example 5.3.

◆ Example 5.5

We will find the variation of the functional

$$v(x_1, x_2) = v(x_1(t), x_2(t)) = \int_0^1 x_1 e^{-x_2} dt,$$

using the formula

$$\delta v = \left. \frac{d}{d\alpha} v(\mathbf{x} + \alpha \delta \mathbf{x}) \right|_{\alpha=0}.$$

We have

$$\begin{aligned} \delta v &= \left. \frac{d}{d\alpha} \left(\int_0^1 (x_1 + \alpha \delta x_1) e^{-(x_2 + \alpha \delta x_2)} dt \right) \right|_{\alpha=0} \\ &= \int_0^1 \left. \frac{d}{d\alpha} ((x_1 + \alpha \delta x_1) e^{-(x_2 + \alpha \delta x_2)}) \right|_{\alpha=0} dt \\ &= \int_0^1 (\delta x_1 e^{-x_2} - \delta x_2 e^{-x_2} (x_1 + \alpha \delta x_1)) dt \Big|_{\alpha=0} \\ &= \int_0^1 (e^{-x_2} \delta x_1 - x_1 e^{-x_2} \delta x_2) dt. \end{aligned}$$

Thus,

$$\delta v = \int_0^1 (e^{-x_2} \delta x_1 - x_1 e^{-x_2} \delta x_2) dt.$$

Definition 5.1 A functional v is maximal on some function x_0 if the value of the functional on any function x close to x_0 is less than or equal to that on x_0 , that is,

$$\Delta v = v(x) - v(x_0) \leq 0.$$

We will now give a first order necessary condition for a given functional to achieve a maximum or minimum on a curve.

Theorem 5.1 If for a functional v there exists a variation and the functional reaches a maximum or a minimum on a curve x_0 , then

$$\delta v(x_0) = 0.$$

Proof Suppose that x_0 and δx are fixed. Then

$$v(x_0 + \alpha \delta x) = \Phi(\alpha);$$

that is, $v(x_0 + \alpha \delta x)$ is a function of α that for $\alpha = 0$ achieves its maximum or minimum. This implies that $\frac{d}{d\alpha} \Phi = 0$, or, equivalently,

$$\left. \frac{d}{d\alpha} v(x_0 + \alpha \delta x) \right|_{\alpha=0} = 0.$$

Hence, by Lemma 5.1, we have $\delta v = 0$. Therefore, a functional that achieves its maximum or a minimum on a function x_0 has its variation evaluated at x_0 equal to zero.

The above results will be used in the next section to determine conditions that a curve has to satisfy to be a maximizer or a minimizer, that is, to be an extremal for a given functional.

5.2.2 Euler–Lagrange Equation

In this section, we consider the following optimization problem.

Problem 1 Suppose that we are given two points (t_0, x_0) and (t_1, x_1) in the (t, x) -plane. We wish to find a curve, or trajectory, joining the given points such that the functional

$$v(x) = \int_{t_0}^{t_1} F(t, x, \dot{x}) dt \quad (5.18)$$

along this trajectory can achieve its extremal—that is, maximal or minimal—value. The problem is illustrated in Figure 5.5. In solving Problem 1, we use the following lemma.

Lemma 5.2 Suppose that a function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is continuous on the interval $[t_0, t_1]$. Then,

$$\int_{t_0}^{t_1} \phi(t) \delta x(t) dt = 0$$

if and only if $\phi(t) = 0$ at every point of the interval $[t_0, t_1]$.

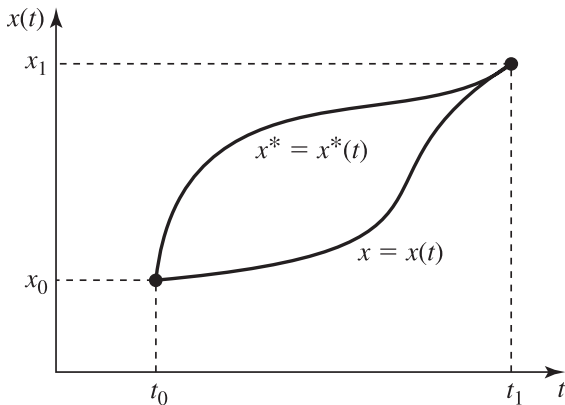


Figure 5.5 Illustration of Problem 1.

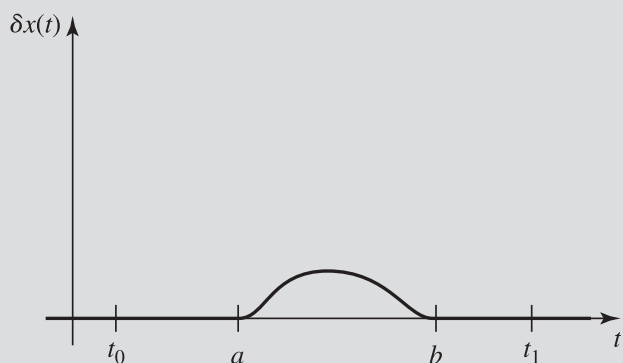


Figure 5.6 Illustration of the proof of Lemma 5.2.

Proof Sufficiency is obvious because we can satisfy $\int_{t_0}^{t_1} \phi(t) \delta x(t) dt = 0$ by choosing ϕ that vanishes at every point of the interval $[t_0, t_1]$.

We prove necessity by contraposition. We thus prove that if ϕ is nonzero on some subinterval of $[t_0, t_1]$, then $\int_{t_0}^{t_1} \phi(t) \delta x(t) dt \neq 0$. Specifically, assume, without loss of generality, that ϕ is positive on the interval $[a, b]$, where $t_0 \leq a \leq b \leq t_1$. Note that if ϕ were nonzero at only one point, then because it is continuous, it would have to be nonzero in some neighborhood of that point. Hence, we can assume that $a < b$. Consider now a variation given by

$$\delta x = \delta x(t) = \begin{cases} k^2(t-a)^2(t-b)^2 & \text{for } a \leq t \leq b, \\ 0 & \text{elsewhere.} \end{cases} \quad (5.19)$$

If we assumed ϕ to be negative on the interval $[a, b]$, then we would use δx to be negative of (5.19). We illustrate δx in Figure 5.6. Because δx is positive in $[a, b]$ and zero elsewhere in the interval $[t_0, t_1]$, the integrand of $\int_{t_0}^{t_1} \phi(t) \delta x(t) dt$ will have the same sign as ϕ in $[a, b]$ and will be zero outside this subinterval. Thus

$$\int_{t_0}^{t_1} \phi(t) \delta x(t) dt = \int_a^b \phi(t) \delta x(t) dt \neq 0,$$

and the proof is complete.

Solution to Problem 1 We assume that there is an optimal trajectory $x = x(t)$ joining the points (t_0, x_0) and (t_1, x_1) such that $\delta v(x(t)) = 0$. Let us then consider an arbitrary acceptable curve $x^* = x^*(t)$ that is close to x and from a family of curves

$$x(t, \alpha) = x(t) + \alpha(x^*(t) - x(t)) = x(t) + \alpha \delta x(t).$$

Note that for $\alpha = 0$ we get $x(t)$, and for $\alpha = 1$ we obtain $x^*(t)$. Furthermore,

$$\frac{d}{dt}(\delta x(t)) = \frac{d}{dt}(x^*(t) - x(t)) = \delta x',$$

and analogously

$$\frac{d^2}{dt^2}(\delta x(t)) = \frac{d^2}{dt^2}(x^*(t) - x(t)) = \delta x''.$$

We are now ready to investigate the behavior of the functional v given by (5.18) on the curves from the family $x(t, \alpha)$. Note that the functional v can be considered as a function of α , that is, $v(x(t, \alpha)) = \Phi(\alpha)$. It follows from the first-order necessary condition for a point to be an extremizer of a function of one variable that $\alpha = 0$ is a candidate to be an extremizer of Φ if

$$\left. \frac{d\Phi(\alpha)}{d\alpha} \right|_{\alpha=0} = 0,$$

where

$$\Phi(\alpha) = \int_{t_0}^{t_1} F(t, x(t, \alpha), x'(t, \alpha)) dt,$$

and $x'(t, \alpha) = \frac{d}{dt}x(t, \alpha)$. We now evaluate $\frac{d\Phi(\alpha)}{d\alpha}$:

$$\frac{d\Phi(\alpha)}{d\alpha} = \int_{t_0}^{t_1} \left(F_x \frac{d}{d\alpha}x(t, \alpha) + F_{x'} \frac{d}{d\alpha}x'(t, \alpha) \right) dt,$$

where $F_x = \frac{\partial}{\partial x}F(t, x(t, \alpha), x'(t, \alpha))$, and $F_{x'} = \frac{\partial}{\partial x'}F(t, x(t, \alpha), x'(t, \alpha))$. Because

$$\frac{d}{d\alpha}x(t, \alpha) = \frac{d}{d\alpha}(x(t) + \alpha\delta x(t)) = \delta x(t)$$

and

$$\frac{d}{d\alpha}x'(t, \alpha) = \frac{d}{d\alpha}(x'(t) + \alpha\delta x'(t)) = \delta x'(t),$$

we can write

$$\frac{d}{d\alpha}\Phi(\alpha) = \int_{t_0}^{t_1} (F_x(t, x(t, \alpha), x'(t, \alpha))\delta x(t) + F_{x'}(t, x(t, \alpha), x'(t, \alpha))\delta x'(t)) dt.$$

By Lemma 5.1,

$$\frac{d}{d\alpha}\Phi(0) = \delta v.$$

Therefore,

$$\delta v = \int_{t_0}^{t_1} (F_x\delta x + F_{x'}\delta x') dt,$$

where for convenience we dropped the argument t . Integrating by parts the last term, along with taking into account that $\delta x' = (\delta x)'$, yields

$$\delta v = (F_{x'}\delta x)|_{t_0}^{t_1} + \int_{t_0}^{t_1} \left(F_x - \frac{d}{dt}F_{x'} \right) \delta x dt.$$

The end points (t_0, x_0) and (t_1, x_1) are fixed. This implies that

$$\delta x|_{t=t_0} = x^*(t_0) - x(t_0) = 0,$$

and

$$\delta x|_{t=t_1} = x^*(t_1) - x(t_1) = 0.$$

Hence

$$\delta v = \int_{t_0}^{t_1} \left(F_x - \frac{d}{dt} F_{x'} \right) \delta x \, dt.$$

By Lemma 5.2,

$$\delta v = 0 \quad \text{if and only if} \quad F_x - \frac{d}{dt} F_{x'} = 0.$$

The equation $F_x - \frac{d}{dt} F_{x'} = 0$ can also be represented as

$$F_x - F_{x't} - F_{x'x}x' - F_{x'x'}x'' = 0.$$

The equation

$$F_x - \frac{d}{dt} F_{x'} = 0 \tag{5.20}$$

is known as the **Euler–Lagrange equation**. The curves $x = x(t, C_1, C_2)$ that are solutions to the Euler–Lagrange equation are called *extremals*. The constants C_1 and C_2 are the integration constants. Thus, a functional can only achieve its extremum on the extremals.

◆ Example 5.6

We find the extremal for the functional

$$v = \int_0^{\pi/2} ((x')^2 - x^2) \, dt, \tag{5.21}$$

where $x(0) = 0$, and $x(\pi/2) = 1$. The Euler–Lagrange equation for the problem takes the form

$$\begin{aligned} F_x - \frac{d}{dt} F_{x'} &= \frac{\partial}{\partial x} ((x')^2 - x^2) - \frac{d}{dt} \left(\frac{\partial}{\partial x'} ((x')^2 - x^2) \right) \\ &= -2x - \frac{d}{dt} (2x') \\ &= -2x - 2x'' \\ &= 0. \end{aligned}$$

Equivalently, the Euler–Lagrange equation for the functional (5.21) can be represented as

$$x'' + x = 0.$$

The characteristic equation of the above differential equation is $s^2 + 1 = 0$.

Therefore, the general solution of the Euler–Lagrange equation has the form

$$x(t) = C_1 \cos t + C_2 \sin t.$$

Taking into account the end-point conditions gives

$$\begin{aligned} x(0) &= C_1 = 0, \\ x\left(\frac{\pi}{2}\right) &= C_2 = 1. \end{aligned}$$

Thus, we have one extremal

$$x(t) = \sin t.$$

The functional (5.21) is minimized or maximized only on the extremal $x(t) = \sin t$.

◆ Example 5.7

We now find the extremal for the functional

$$v = \int_0^1 ((x')^2 + 12xt) dt, \quad (5.22)$$

where $x(0) = 0$, and $x(1) = 1$. The Euler–Lagrange equation for the this functional is

$$\begin{aligned} F_x - \frac{d}{dt} F_{x'} &= \frac{\partial}{\partial x} ((x')^2 + 12xt) - \frac{d}{dt} \left(\frac{\partial}{\partial x'} ((x')^2 + 12xt) \right) \\ &= 12t - \frac{d}{dt} (2x') \\ &= 12t - 2x'' \\ &= 0. \end{aligned}$$

Equivalently, the Euler–Lagrange equation for the functional (5.22) can be represented as

$$x'' = 6t.$$

The general solution to the above differential equation is

$$x(t) = C_1 t + C_2 + t^3.$$

From the end-point conditions, we get

$$\begin{aligned} x(0) &= C_2 = 0, \\ x(1) &= C_1 + 1 = 1. \end{aligned}$$

Hence, $C_1 = 0$, and the extremal has the form

$$x(t) = t^3.$$

Suppose now that $A = A(t_0, x_{10}, \dots, x_{n0})$ and $B = B(t_1, x_{11}, \dots, x_{n1})$ are two given points in the $(n + 1)$ -dimensional space $(t, x_1, \dots, x_n)$, along with the functional

$$v = v(x_1, \dots, x_n) = \int_{t_0}^{t_1} F(t, x_1, \dots, x_n, \dot{x}_1, \dots, \dot{x}_n) dt.$$

To find a trajectory along which $\delta v = 0$, we consider only the variation in one coordinate, say x_i , while other coordinates are fixed. We perform this process for each coordinate. The functional v will depend only upon x_i , and we write

$$v(x_1, \dots, x_i, \dots, x_n) = \tilde{v}(x_i).$$

Thus, we reduced the given n -dimensional problem to n one-dimensional problems. Proceeding as before, we conclude that the functional $v = v(x_1, \dots, x_n)$ can only be extremized on the trajectories that satisfy **n Euler–Lagrange equations**,

$$F_{x_i} - \frac{d}{dt} F_{\dot{x}_i} = 0, \quad i = 1, \dots, n. \quad (5.23)$$

Observe that if in (5.23) we set **$F = L = K - U$** , where L is the Lagrangian, $K = K(\dot{q}_1, \dots, \dot{q}_n)$ is the kinetic energy of the particle, $U = U(q_1, \dots, q_n)$ is its potential energy, and $x_i = q_i$, then (5.23) becomes

$$\frac{\partial L}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = 0, \quad i = 1, 2, \dots, n,$$

which are the **Lagrange equations of motion for a particle**. Thus a particle moves along a path on which the functional

$$J = \int_{t_1}^{t_2} L dt$$

is extremized, which agrees with the Hamilton principle saying that a particle moves along the path that minimizes the integral J .

In our discussion up to now, both end points were fixed. We now would like to solve a more general optimization problem, where the end points need not be fixed. For simplicity, we assume first that one of the end points is fixed, while the other is free. After we solve this problem, we can easily generalize to the case when both end points are free.

Problem 2 In the (t, x) plane the left end point A is fixed, while the right end point B is free (see Figure 5.7). It is required to find a trajectory $x = x(t)$ starting at A on which the functional $v = \int_{t_0}^{t_1} F(t, x, x') dt$ can achieve its extremal value.

Solution to Problem 2 Following Elsgolc [75, pp. 64–72], we begin by assuming that a solution to Problem 2 exists; that is, there is a trajectory $x = x(t)$ joining (t_0, x_0) and (t_1, x_1) such that $\delta v = 0$. This trajectory must satisfy the Euler–Lagrange equation. Consider an adjoining

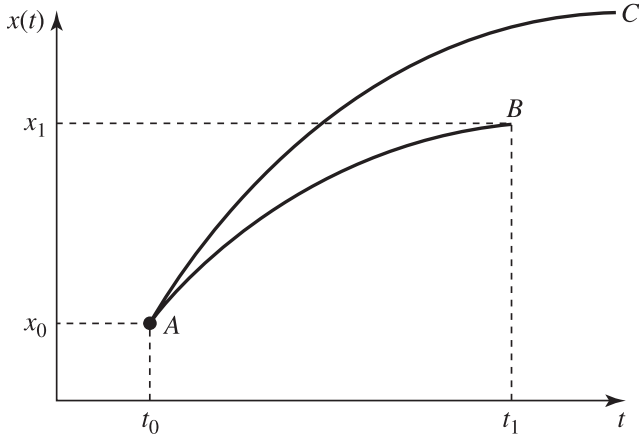


Figure 5.7 Illustration of Problem 2.

trajectory specified by $x + \delta x$ joining (t_0, x_0) and $(t_1 + \delta t_1, x_1 + \delta x_1)$. Then,

$$\begin{aligned} \Delta v &= v(x + \delta x) - v(x) \\ &= \int_{t_0}^{t_1 + \delta t_1} F(t, x + \delta x, x' + \delta x') dt - \int_{t_0}^{t_1} F(t, x, x') dt \\ &= \int_{t_1}^{t_1 + \delta t_1} F(t, x + \delta x, x' + \delta x') dt + \int_{t_0}^{t_1} (F(t, x + \delta x, x' + \delta x') - F(t, x, x')) dt. \end{aligned}$$

Our goal now is to extract from Δv its linear portion in δx , which we know is the variation of the given functional. We begin by approximating the first integral as

$$\int_{t_1}^{t_1 + \delta t_1} F(t, x + \delta x, x' + \delta x') dt \approx F(t, x, x')|_{t=t_1} \delta t_1.$$

Expanding the integrand of the second integral into a Taylor series and retaining only first-order terms yields

$$\int_{t_0}^{t_1} (F(t, x + \delta x, x' + \delta x') - F(t, x, x')) dt = \int_{t_0}^{t_1} (F_x(t, x, x')\delta x + F_{x'}(t, x, x')\delta x') dt.$$

Integrating the second term by parts and combining with the first term gives

$$\int_{t_0}^{t_1} (F_x\delta x + F_{x'}\delta x') dt = (F_{x'}\delta x)|_{t_0}^{t_1} + \int_{t_0}^{t_1} \left(F_x - \frac{d}{dt} F_{x'} \right) \delta x dt,$$

where for simplicity of notation we suppressed the arguments of F_x and $F_{x'}$. We assumed that a solution to the problem exists. Hence, an extremal must satisfy the Euler–Lagrange equation $F_x - \frac{d}{dt} F_{x'} = 0$. Because the left end point is fixed, we obtain

$$\delta x|_{t=t_0} = 0.$$

Taking into account the results of the above manipulations yields

$$\Delta v \approx F(t, x, x')|_{t=t_1} \delta t_1 + (F_{x'}\delta x)|_{t=t_1}. \quad (5.24)$$

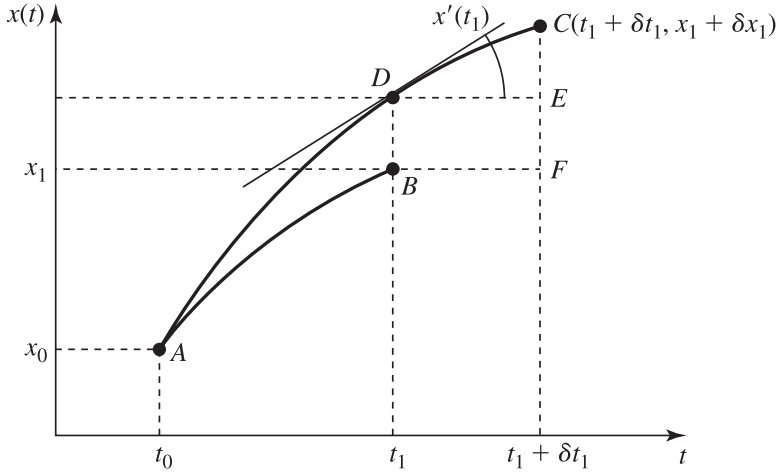


Figure 5.8 A geometric interpretation of calculating the variation of the functional for Problem 2.

The final step in obtaining δv is to evaluate $(F_{x'}\delta x)|_{t_1}$. The problem is two-dimensional, so we can illustrate it graphically as shown in Figure 5.8. Referring to Figure 5.8, we obtain

$$\begin{aligned} DB &= \delta x|_{t=t_1}, \\ FC &= \delta x_1, \\ EC &\approx x'(t_1)\delta t_1, \\ DB &= FC - EC. \end{aligned}$$

Hence,

$$\delta x|_{t=t_1} \approx \delta x_1 - x'(t_1)\delta t_1.$$

Substituting the above into (5.24) gives

$$\begin{aligned} \delta v &= F(t, x, x')|_{t=t_1} \delta t_1 + F_{x'}|_{t=t_1} (\delta x_1 - x'(t_1)\delta t_1) \\ &= (F - x'F_{x'})|_{t=t_1} \delta t_1 + F_{x'}|_{t=t_1} \delta x_1. \end{aligned}$$

If the variations δt_1 and δx_1 are arbitrary, we obtain $\delta v = 0$ if in addition to the Euler–Lagrange equation the following conditions are also satisfied:

$$(F - x'F_{x'})|_{t=t_1} = 0 \quad \text{and} \quad F_{x'}|_{t=t_1} = 0.$$

The above end-point relations are called the *transversality conditions*. In conclusion, the optimal trajectory must satisfy both the Euler–Lagrange equation and the transversality conditions.

◆ Example 5.8

Consider the functional of Example 5.7, where now the left end point is fixed as before—that is, $x(0) = 0$ —while the right end point $x(1)$ is free. Here, we have $t_1 = 1$. Because $\delta t_1 = 0$ and δx_1 is arbitrary, the transversality conditions for the right end point are reduced to one condition,

$$F_{x'}|_{t=t_1} = 0.$$

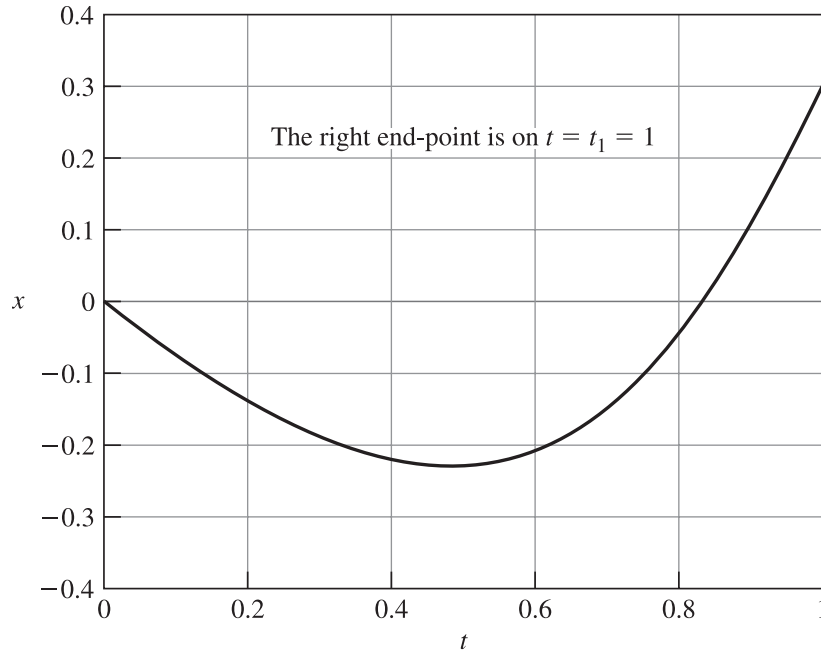


Figure 5.9 A plot of the extremal, $x(t) = -3t + t^3$, of Example 5.8.

Taking into account the solution to the Euler–Lagrange equation obtained in Example 5.7 and then using the fact that the left end point is fixed, we obtain

$$\begin{aligned}
 0 &= F_{x'}|_{t=t_1} \\
 &= 2x'(t_1) \\
 &= 2(C_1 + 3t^2)|_{t=1} \\
 &= 2(C_1 + 3).
 \end{aligned}$$

Hence, $C_1 = -3$ and the extremal for this case is

$$x(t) = -3t + t^3.$$

In Figure 5.9, we show a plot of the above extremal.

◆ Example 5.9

We consider again the same functional as in Example 5.7, where, this time, the left end point is fixed, as before, while at the right end point t_1 is free and $x(t_1) = 1$. Hence, in this case, $\delta x_1 = 0$ while δt_1 is arbitrary. The transversality conditions reduce to one condition,

$$(F - x'F_{x'})|_{t=t_1} = 0.$$

We perform simple calculations to obtain

$$F - x'F_{x'} = 12xt - (x')^2,$$

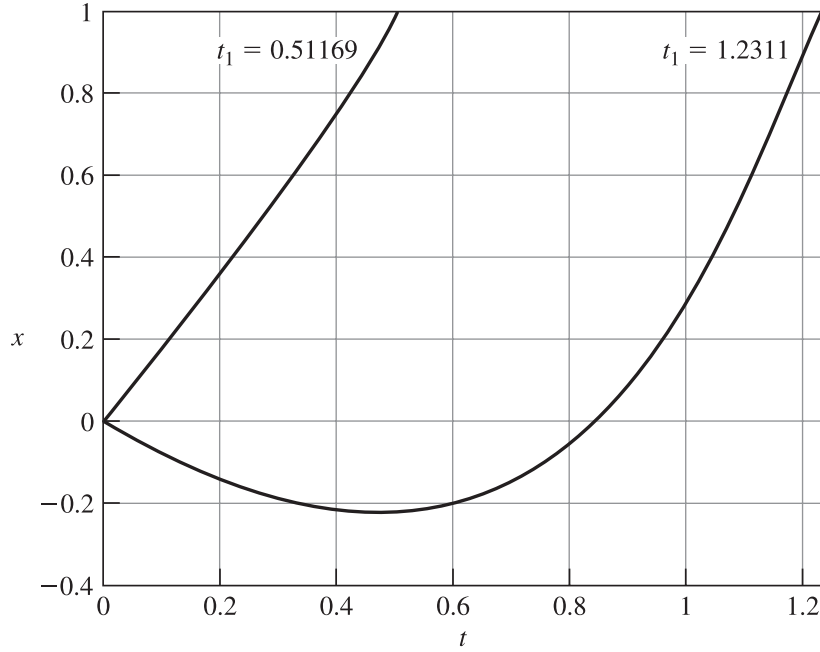


Figure 5.10 Plots of the extremals of Example 5.9.

where

$$x' = (C_1 t + t^3)' = C_1 + 3t^2.$$

Note that because $x(t_1) = C_1 t_1 + t_1^3 = 1$, we have

$$C_1 = \frac{1 - t_1^3}{t_1}.$$

Taking into account the above and the fact that $x(t_1) = 1$, we perform simple algebraic manipulations to arrive at the following equation for t_1 :

$$(F - x'F_{x'})|_{t=t_1} = (12x(t_1)t_1 - (x'(t_1))^2) = -4t_1^6 + 8t_1^3 - 1 = 0.$$

Solving the above equation, we obtain the following two solutions:

$$t_1 = \left(1 \pm \sqrt{\frac{3}{4}}\right)^{1/3}.$$

Plots of the two extremals for this example are shown in Figure 5.10.

If the position of the end point B is restricted to a curve $x = g_1(t)$, then

$$\delta x_1 \approx g_1'(t_1)\delta t_1,$$

and hence

$$\delta v = (F + (g_1' - x')F_{x'})\delta t_1 = 0$$

if

$$(F + (g'_1 - x')F_{x'})|_{t=t_1} = 0. \quad (5.25)$$

The above is the transversality condition for the case when the end point B is restricted to the curve $x = g_1(t)$.

In the next section we discuss a specific problem of minimizing a functional subject to constraints in the form of differential equations.

5.3 Linear Quadratic Regulator

5.3.1 Algebraic Riccati Equation (ARE)

Consider a linear system modeled by

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \quad \mathbf{x}(0) = \mathbf{x}_0$$

and the associated performance index

$$J = \int_0^\infty (\mathbf{x}^T \mathbf{Q}\mathbf{x} + \mathbf{u}^T \mathbf{R}\mathbf{u}) dt,$$

where $\mathbf{Q} = \mathbf{Q}^T \geq 0$ and $\mathbf{R} = \mathbf{R}^T > 0$. Our goal is to construct a stabilizing linear state-feedback controller of the form $\mathbf{u} = -\mathbf{K}\mathbf{x}$ that minimizes the performance index J . We denote such a linear control law by $\mathbf{u}^*$.

First, we assume that a linear state-feedback optimal controller exists such that the optimal closed-loop system

$$\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{B}\mathbf{K})\mathbf{x}$$

is asymptotically stable. The above assumption implies that there is a Lyapunov function $V = \mathbf{x}^T \mathbf{P}\mathbf{x}$ for the closed-loop system; that is, for some positive definite matrix $\mathbf{P}$ the time derivative dV/dt evaluated on the trajectories of the closed-loop system is negative definite.

Theorem 5.2 If the state-feedback controller $\mathbf{u}^* = -\mathbf{K}\mathbf{x}$ is such that

$$\min_{\mathbf{u}} \left(\frac{dV}{dt} + \mathbf{x}^T \mathbf{Q}\mathbf{x} + \mathbf{u}^T \mathbf{R}\mathbf{u} \right) = 0, \quad (5.26)$$

for some $V = \mathbf{x}^T \mathbf{P}\mathbf{x}$, then the controller is optimal.

Proof We can rewrite the condition of the theorem as

$$\left. \frac{dV}{dt} \right|_{\mathbf{u}=\mathbf{u}^*} + \mathbf{x}^T \mathbf{Q}\mathbf{x} + \mathbf{u}^{*T} \mathbf{R}\mathbf{u}^* = 0.$$

Hence,

$$\left. \frac{dV}{dt} \right|_{\mathbf{u}=\mathbf{u}^*} = -\mathbf{x}^T \mathbf{Q}\mathbf{x} - \mathbf{u}^{*T} \mathbf{R}\mathbf{u}^*.$$

Integrating both sides of the resulting equation with respect to time from 0 to ∞ , we obtain

$$V(\mathbf{x}(\infty)) - V(\mathbf{x}(0)) = - \int_0^\infty (\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^{*T} \mathbf{R} \mathbf{u}^*) dt.$$

Because by assumption the closed-loop system is asymptotically stable, $\mathbf{x}(\infty) = \mathbf{0}$, and therefore

$$V(\mathbf{x}(0)) = \mathbf{x}_0^T \mathbf{P} \mathbf{x}_0 = \int_0^\infty (\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^{*T} \mathbf{R} \mathbf{u}^*) dt.$$

Thus, we showed that if a linear state-feedback controller satisfies the assumption of the theorem, then the value of the performance index for such a controller is

$$J(\mathbf{u}^*) = \mathbf{x}_0^T \mathbf{P} \mathbf{x}_0.$$

To show that such a controller is indeed optimal, we use a proof by contradiction. We assume that (5.26) holds and that $\mathbf{u}^*$ is not optimal. Suppose that $\tilde{\mathbf{u}}$ yields a smaller value of J , that is,

$$J(\tilde{\mathbf{u}}) < J(\mathbf{u}^*).$$

It follows from (5.26) that

$$\left. \frac{dV}{dt} \right|_{\mathbf{u}=\tilde{\mathbf{u}}} + \mathbf{x}^T \mathbf{Q} \mathbf{x} + \tilde{\mathbf{u}}^T \mathbf{R} \tilde{\mathbf{u}} \geq 0,$$

that is,

$$\left. \frac{dV}{dt} \right|_{\mathbf{u}=\tilde{\mathbf{u}}} \geq -\mathbf{x}^T \mathbf{Q} \mathbf{x} - \tilde{\mathbf{u}}^T \mathbf{R} \tilde{\mathbf{u}}.$$

Integrating the above with respect to time from 0 to ∞ yields

$$V(\mathbf{x}(0)) \leq \int_0^\infty (\mathbf{x}^T \mathbf{Q} \mathbf{x} + \tilde{\mathbf{u}}^T \mathbf{R} \tilde{\mathbf{u}}) dt$$

implying that

$$J(\mathbf{u}^*) \leq J(\tilde{\mathbf{u}}),$$

which is a contradiction, and the proof is complete.

It follows from the above theorem that the synthesis of the optimal control law involves finding an appropriate Lyapunov function, or equivalently, the matrix $\mathbf{P}$. The appropriate $\mathbf{P}$ is found by minimizing

$$f(\mathbf{u}) = \frac{dV}{dt} + \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u}, \quad (5.27)$$

We first apply to (5.27) the necessary condition for unconstrained minimization,

$$\frac{\partial}{\partial \mathbf{u}} \left(\frac{dV}{dt} + \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u} \right) \bigg|_{\mathbf{u}=\mathbf{u}^*} = \mathbf{0}^T.$$

Differentiating the above yields

$$\begin{aligned}
 \frac{\partial}{\partial \mathbf{u}} \left(\frac{dV}{dt} + \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u} \right) &= \frac{\partial}{\partial \mathbf{u}} (2\mathbf{x}^T \mathbf{P} \dot{\mathbf{x}} + \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u}) \\
 &= \frac{\partial}{\partial \mathbf{u}} (2\mathbf{x}^T \mathbf{P} \mathbf{A} \mathbf{x} + 2\mathbf{x}^T \mathbf{P} \mathbf{B} \mathbf{u} + \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u}) \\
 &= 2\mathbf{x}^T \mathbf{P} \mathbf{B} + 2\mathbf{u}^T \mathbf{R} \\
 &= \mathbf{0}^T.
 \end{aligned}$$

Hence, the candidate for an optimal control law has the form

$$\mathbf{u}^* = -\mathbf{R}^{-1} \mathbf{B}^T \mathbf{P} \mathbf{x} = -\mathbf{K} \mathbf{x}, \quad (5.28)$$

where $\mathbf{K} = \mathbf{R}^{-1} \mathbf{B}^T \mathbf{P}$. Note that

$$\begin{aligned}
 \frac{\partial^2}{\partial \mathbf{u}^2} \left(\frac{dV}{dt} + \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u} \right) &= \frac{\partial^2}{\partial \mathbf{u}^2} (2\mathbf{x}^T \mathbf{P} \mathbf{A} \mathbf{x} + 2\mathbf{x}^T \mathbf{P} \mathbf{B} \mathbf{u} + \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u}) \\
 &= \frac{\partial}{\partial \mathbf{u}} (2\mathbf{x}^T \mathbf{P} \mathbf{B} + 2\mathbf{u}^T \mathbf{R}) \\
 &= 2\mathbf{R} \\
 &> 0.
 \end{aligned}$$

Thus the second-order sufficiency condition for $\mathbf{u}^*$ to minimize (5.27) is satisfied.

We now turn our attention to finding an appropriate matrix $\mathbf{P}$. The optimal closed-loop system has the form

$$\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T \mathbf{P}) \mathbf{x}, \quad \mathbf{x}(0) = \mathbf{x}_0.$$

Our optimal controller satisfies the equation

$$\left. \frac{dV}{dt} \right|_{\mathbf{u}=\mathbf{u}^*} + \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^{*T} \mathbf{R} \mathbf{u}^* = 0,$$

that is,

$$2\mathbf{x}^T \mathbf{P} \mathbf{A} \mathbf{x} + 2\mathbf{x}^T \mathbf{P} \mathbf{B} \mathbf{u}^* + \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^{*T} \mathbf{R} \mathbf{u}^* = 0.$$

We substitute the expression for $\mathbf{u}^*$, given by (5.28), into the above equation and represent it as

$$\mathbf{x}^T (\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A}) \mathbf{x} - 2\mathbf{x}^T \mathbf{P} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T \mathbf{P} \mathbf{x} + \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{x}^T \mathbf{P} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T \mathbf{P} \mathbf{x} = 0.$$

Factoring out $\mathbf{x}$ yields

$$\mathbf{x}^T (\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} + \mathbf{Q} - \mathbf{P} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T \mathbf{P}) \mathbf{x} = 0.$$

The above equation should hold for any $\mathbf{x}$. For this to be true we have to have

$$\boxed{\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} + \mathbf{Q} - \mathbf{P} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T \mathbf{P} = \mathbf{O}.} \quad (5.29)$$

The above equation is referred to as the *algebraic Riccati equation (ARE)*. In conclusion, the synthesis of the optimal linear state feedback controller minimizing the performance index

$$J = \int_0^\infty (\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u}) dt$$

subject to

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \quad \mathbf{x}(0) = \mathbf{x}_0$$

requires solving the ARE given by (5.29).

◆ Example 5.10

Consider the following model of a dynamical system:

$$\dot{x} = 2u_1 + 2u_2, \quad x(0) = 3,$$

along with the associated performance index

$$J = \int_0^\infty (x^2 + ru_1^2 + ru_2^2) dt,$$

where $r > 0$ is a parameter.

1. We first find the solution to the ARE corresponding to the linear state-feedback optimal controller. We have

$$\mathbf{A} = 0, \quad \mathbf{B} = [2 \ 2], \quad \mathbf{Q} = 1, \quad \mathbf{R} = r \mathbf{I}_2.$$

The ARE for this problem is

$$\mathbf{O} = \mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} + \mathbf{Q} - \mathbf{P} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T \mathbf{P} = 1 - \frac{8}{r} p^2,$$

whose solution is

$$p = \sqrt{\frac{r}{8}}.$$

2. We now write the equation of the closed-loop system driven by the optimal controller. The optimal controller has the form

$$\mathbf{u} = -\mathbf{R}^{-1} \mathbf{B}^T \mathbf{P} \mathbf{x} = -\frac{1}{\sqrt{2r}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} x.$$

Hence, the closed-loop optimal system is described by

$$\dot{x} = [2 \ 2] \mathbf{u} = -\frac{4}{\sqrt{2r}} x.$$

3. Finally, we find the value of J for the optimal closed-loop system. We have

$$J = \mathbf{x}^T(0) \mathbf{P} \mathbf{x}(0) = \frac{9}{2} \sqrt{\frac{r}{2}}.$$

Thus, we have three candidate solutions to the problem:

$$\begin{aligned} u^1 &= -x_1 - x_2, \\ u^2 &= -x_1 + 3x_2, \\ u^3 &= x_1 + x_2. \end{aligned}$$

However, only u^1 stabilizes the plant. Hence, the optimal control law for the problem is

$$u^* = -x_1 - x_2.$$

Note that only for this controller the boundary condition, $J^*(\mathbf{x}(t_f)) = J^*(\mathbf{x}(\infty)) = 0$, is satisfied.

5.5 Pontryagin's Minimum Principle

5.5.1 Optimal Control with Constraints on Inputs

In this section we provide an informal derivation of the minimum principle of Pontryagin, which is a generalization of the Euler–Lagrange equations that also includes problems with constraints on the control inputs. Only a special case of the minimum principle will be stated, but this special case covers a large class of control problems. We consider the problem of minimizing the performance index given by

$$J = \Phi(t_f, \mathbf{x}(t_f)) + \int_{t_0}^{t_f} F(\mathbf{x}(t), \mathbf{u}(t)) dt \quad (5.87)$$

subject to

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}), \quad \mathbf{x}(t_0) = \mathbf{x}_0, \quad \text{and} \quad \mathbf{x}(t_f) = \mathbf{x}_f. \quad (5.88)$$

We consider two cases: fixed final state and free final state.

It follows from our discussion of the HJB equation that the optimal controller, $\mathbf{u}^*$, minimizes the Hamiltonian function, that is,

$$\mathbf{u}^* = \arg \min_{\mathbf{u} \in U} H(\mathbf{x}, \mathbf{u}, \mathbf{p}),$$

where $H(\mathbf{x}, \mathbf{u}, \mathbf{p}) = H = F + \mathbf{p}^T \mathbf{f}$, and

$$\mathbf{p} = [p_1 \quad p_2 \quad \cdots \quad p_n]^T = \left[\frac{\partial J^*}{\partial x_1} \quad \frac{\partial J^*}{\partial x_2} \quad \cdots \quad \frac{\partial J^*}{\partial x_n} \right]^T.$$

To proceed, we need one more assumption in addition to the assumptions we made while discussing the HJB equation. We assume that $J^*(\mathbf{x})$ has continuous second partial derivatives $\partial^2 J^* / \partial x_i \partial x_j$; that is, the functions p_i have continuous first derivatives $\partial p_i / \partial x_j$. This implies that

$$\frac{\partial \mathbf{p}}{\partial \mathbf{x}} = \left(\frac{\partial \mathbf{p}}{\partial \mathbf{x}} \right)^T. \quad (5.89)$$

We also assume that the components of $\mathbf{f}$ have continuous first derivatives $\partial f_i / \partial x_j$. We then

compute

$$\begin{aligned}\frac{\partial H}{\partial \mathbf{x}} &= \frac{\partial}{\partial \mathbf{x}}(F + \mathbf{p}^T \mathbf{f}) \\ &= \frac{\partial F}{\partial \mathbf{x}} + \mathbf{f}^T \frac{\partial \mathbf{p}}{\partial \mathbf{x}} + \mathbf{p}^T \frac{\partial \mathbf{f}}{\partial \mathbf{x}},\end{aligned}\quad (5.90)$$

where $\partial F / \partial \mathbf{x} = \nabla_{\mathbf{x}}^T F$. From our previous discussion it follows that the Hamiltonian function attains its minimum on the optimal trajectory. Taking this into account, we conclude that for the optimal process we obtain

$$\frac{\partial H}{\partial \mathbf{x}} = \mathbf{0}^T \quad \text{for } (\mathbf{x}, \mathbf{u}) = (\mathbf{x}^*, \mathbf{u}^*).$$

Hence, for the optimal process $(\mathbf{x}^*, \mathbf{u}^*)$ we have

$$\frac{\partial F}{\partial \mathbf{x}} + \mathbf{f}^T \frac{\partial \mathbf{p}}{\partial \mathbf{x}} + \mathbf{p}^T \frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \mathbf{0}^T. \quad (5.91)$$

We now differentiate $\mathbf{p}$ with respect to time to get

$$\dot{\mathbf{p}} = \frac{\partial \mathbf{p}}{\partial \mathbf{x}} \dot{\mathbf{x}} = \frac{\partial \mathbf{p}}{\partial \mathbf{x}} \mathbf{f}. \quad (5.92)$$

Combining (5.91) and (5.92) and taking into account (5.89), we obtain

$$\dot{\mathbf{p}} = - \left(\frac{\partial F}{\partial \mathbf{x}} \right)^T - \left(\frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right)^T \mathbf{p}. \quad (5.93)$$

Thus, $\mathbf{p}$ is the solution to the differential equation (5.93), and hence $\mathbf{p} = \mathbf{p}(t)$. Therefore $\partial \mathbf{p} / \partial \mathbf{x} = \mathbf{0}$, and by (5.90) we have

$$\frac{\partial H}{\partial \mathbf{x}} = \frac{\partial F}{\partial \mathbf{x}} + \mathbf{p}^T \frac{\partial \mathbf{f}}{\partial \mathbf{x}}. \quad (5.94)$$

Comparing (5.93) and (5.94) gives

$$\dot{\mathbf{p}} = - \left(\frac{\partial H}{\partial \mathbf{x}} \right)^T \quad (5.95)$$

The above equation is called in the literature the *adjoint* or *costate* equation.

We summarize our development in the following theorem.

Theorem 5.6 Necessary conditions for $\mathbf{u} \in U$ to minimize (5.87) subject to (5.88) are

$$\dot{\mathbf{p}} = - \left(\frac{\partial H}{\partial \mathbf{x}} \right)^T,$$

where $H = H(\mathbf{x}, \mathbf{u}, \mathbf{p}) = F(\mathbf{x}, \mathbf{u}) + \mathbf{p}^T \mathbf{f}(\mathbf{x}, \mathbf{u})$, and

$$H(\mathbf{x}^*, \mathbf{u}^*, \mathbf{p}^*) = \min_{\mathbf{u} \in U} H(\mathbf{x}, \mathbf{u}, \mathbf{p}).$$

If the final state, $\mathbf{x}(t_f)$, is free, then in addition to the above conditions it is required that the following end-point condition is satisfied:

$$\mathbf{p}(t_f) = \nabla_{\mathbf{x}} \Phi|_{t=t_f}.$$

We now illustrate the above theorem with the following well-known example.

◆ Example 5.21

We consider a controlled object modeled by

$$\begin{aligned} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \\ &= \mathbf{A}\mathbf{x} + \mathbf{b}u \end{aligned}$$

with the performance index

$$J = \int_0^{t_f} dt.$$

The control is required to satisfy

$$|u(t)| \leq 1$$

for all $t \in [0, t_f]$. This constraint means that the control must have magnitude no greater than 1. Our objective is to find admissible control that minimizes J and transfers the system from a given initial state $\mathbf{x}_0$ to the origin.

We begin by finding the Hamiltonian function for the problem:

$$\begin{aligned} H &= 1 + \mathbf{p}^T (\mathbf{A}\mathbf{x} + \mathbf{b}u) \\ &= 1 + [p_1 \quad p_2] \left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \right) \\ &= 1 + p_1 x_2 + p_2 u. \end{aligned}$$

The costate equations are

$$\begin{bmatrix} \dot{p}_1 \\ \dot{p}_2 \end{bmatrix} = - \begin{bmatrix} \frac{\partial H}{\partial x_1} \\ \frac{\partial H}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 0 \\ -p_1 \end{bmatrix}.$$

Solving the costate equations yields

$$p_1 = d_1 \quad \text{and} \quad p_2 = -d_1 t + d_2,$$

where d_1 and d_2 are integration constants. We now find an admissible control minimizing the Hamiltonian:

$$\begin{aligned} \arg_u \min H &= \arg_u \min (1 + p_1 x_2 + p_2 u) \\ &= \arg_u \min (p_2 u) \\ &= \begin{cases} u(t) = 1 & \text{if } p_2 < 0, \\ u(t) = -1 & \text{if } p_2 > 0. \end{cases} \end{aligned}$$

Hence,

$$u^*(t) = -\text{sign}(p_2^*) = -\text{sign}(-d_1 t + d_2),$$

where

$$\text{sign}(z) = \begin{cases} 1 & \text{if } z > 0, \\ -1 & \text{if } z < 0. \end{cases}$$